Scenario 2 — A Comparative Study of Prompt Adherence in OpenAI and Gemini Image Generation

CET946 Assignment 3 — GenAI Image Generation Case Study

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Study type: Individual, comparative image-generation experiment

Tools compared: OpenAI image-generation API and Google Gemini image-generation API

Source code and project repository: https://github.com/mlnotes2718/image_gen_study

Executive Summary....3
1. Introduction and Selected Scenario....3
2. Tools and Personal Workflow....4
2.1 Ease of use, learning curve, and trade-offs....4
3. Experimental Design....5
3.1 Test objectives and increasing task demands....5
3.2 Controls and procedure....5
3.3 Scoring rubric....6
4. Quantitative Results....6
4.1 Detailed interpretation of Test 2....6
5. Visual Comparison and Discussion....7
5.1 T01 — Object count, shape, and color....7
5.2 T02 — Spatial composition versus geometric integrity....7
5.3 T03 — Exact text versus camera viewpoint....8
5.4 T04 — Complex scene, lighting, and aesthetics....8
5.5 Style, aesthetics, realism, and visual quality....8
6. Strengths and Weaknesses of Each Tool....9
7. Preferred Tool and Intended Future Use....9
8. Missing Capabilities and Desired Improvements....9
9. Limitations of the Evaluation....10
10. Conclusion....10

Appendix A

T01 - Counting and Object Generation....12
Provider: OpenAI....12
Provider: Gemini....15
T02 - Spatial Relationship between objects....18
Provider: OpenAI....19
Provider: Gemini....22
T03 - Generate Image with Text Information....25
Provider: OpenAI....26
Provider: Gemini....29
T04 - Generate Complex Image....32
Provider: OpenAI....33
Provider: Gemini....36

Executive Summary

This study compares two generative AI image-generation tools through a controlled set of text-to-image tasks. OpenAI's `gpt-image-2.5-sunburst` and Google's `gemini-3-pro-image` were accessed using Python in a Jupyter notebook. Four prompts tested object counting, unusual spatial relationships, accurate text rendering, and complex photorealistic scene composition. Each prompt was generated three times per model, producing **24 images** in total. The same prompt was used for both models within each test.

Both tools performed strongly on the explicitly scored requirements. Across the four tests, OpenAI scored **196/198 (99.0%)**, while Gemini scored **198/198 (100%)**. The two-point difference came from two OpenAI images in Test 2, where an unnatural pointed feature appeared at the contact between a cube and a pyramid. These were classified as **shape-preservation** errors, not failures of the intended spatial relationship. The results do not establish a general ranking of the two tools: the sample is small, the tasks are limited, and photographic realism was evaluated qualitatively rather than through the numerical rubric.

The main lesson is that satisfying a prompt's visible content is not identical to preserving convincing object geometry, camera viewpoint, or physical realism. For this particular study, **Gemini (Nano Banana Pro) is the provisional preferred tool** because it preserved the intended cube geometry more consistently while matching OpenAI's scores on the other tests. This preference is specific to the tested prompts and should not be considered as general product recommendation.

1. Introduction and Selected Scenario

This report addresses **Scenario 2**, which requires a comparison of two generative AI image-generation tools, including usability, visual quality, prompt complexity, a preferred tool, and desired future improvements. This study focuses on **text-to-image generation**, not image editing.

The central research question is: **How accurately do two image-generation models follow increasingly demanding text prompts, and where do differences emerge beyond the numerical score?** Prompt adherence is assessed through clearly specified objects, relationships, text, and scene elements. Photographic realism and aesthetic qualities are assessed separately to avoid conflating visual appeal with instruction-following accuracy.

2. Tools and Personal Workflow

Provider	OpenAI	Google Gemini
Model used	`gpt-image-2.5-sunburst`	`gemini-3-pro-image`
Access method	OpenAI API from Python/Jupyter notebook	Gemini API from Python/Jupyter notebook
Output evaluated	Generated images	Generated images
Repetitions per prompt	3	3

The study used APIs rather than consumer web interfaces because the API workflow allowed us to specify the model used and run a repeatable set of prompts. The image-generation code was AI-assisted and then manually reviewed. Source code, images, and the study materials are linked in the project repository.

2.1 Ease of use, learning curve, and trade-offs

Ease of Use: Using the web interface is easy. The challenge will be coming out with the prompt. However, there is a constraint, we do not know what model is used to generate such an image. There is no venue for comparison unless we use the API.

Using API: A Jupyter notebook provides a convenient environment for keeping prompts, model calls, image files, and scoring together. It also supports repeatable comparisons without manually retyping prompts into separate web interfaces. This is particularly useful for someone comfortable with Python.

Learning Curve and Setup: Compared with a conversational web interface, API experimentation requires environment setup, credentials, Python code, handling generated image responses, and organizing outputs. These add a technical learning curve and make the workflow less immediately accessible to non-programmers. API use may also involve usage charges.

Trade-Off: For image generation in creative settings, the web interface is good enough. However, if we need more control on which model to use, using API provides better control on which model will be used. Using API allows us to perform cross platform generation and comparison.

3. Experimental Design

3.1 Test objectives and increasing task demands

Test	Task	Primary evaluated capability	Relative task demand
T01	Nine colored geometric objects	Counting, shape, and color	Simple
T02	Four objects in an unusual arrangement	Shape preservation, color, and spatial relationships	Moderate
T03	Airport departure board	Exact reproduction of structured text	High text precision
T04	Minimalist beachside living room	Multiple furniture, spatial, lighting, and background requirements	High scene complexity

The tasks move from a relatively simple object inventory to prompts involving spatial geometry, exact text, and integrated scenes.

3.2 Controls and procedure

1. Use the same text prompt for both models in each test.
2. Generate three independent images per model per test, yielding $4 \times 2 \times 3 =$ **24 images**.
3. Inspect each generated image manually against the prompt.
4. Record quantitative scores using predefined criteria and document qualitative observations separately.
5. Preserve the generated images and detailed scores in Appendix A.

3.3 Scoring rubric

Test	Scored items per image	Maximum
T01	Shape of each of 9 objects (9) and color of each (9)	18
T02	Shape of each of 4 objects (4), color (4), and requested spatial placement (4)	12
T03	Four column headings plus 24 flight data fields (6 rows × 4 fields)	28
T04	Sofa, coffee table, armchair pair, floor lamp, plant, sliding door, lighting, outdoor background	8

Each criterion receives one point when judged satisfied. In T04, each scored item is a bundled requirement: for example, the sliding-door item includes its requested location and open state, although the images do not always make the latter easy to verify. Accordingly, the reported T04 scores should be interpreted as the evaluator's overall judgments rather than independently verified satisfaction of every fine-grained prompt attribute.

Some instructions—including absence of extra decorations, precise camera angle, perfect reflection physics, and fine details of furniture placement—are discussed qualitatively and do not all receive separate numerical points. **A perfect numerical score therefore does not mean that every sentence of the prompt was verified as perfectly satisfied.**

4. Quantitative Results

Test	OpenAI	Gemini	Interpretation
T01 — Counting, shape, color	54/54	54/54	Both met the object-level rubric
T02 — Shape, color, spatial	34/36	36/36	Two OpenAI shape deductions
T03 — Text accuracy	84/84	84/84	All scored text fields correct
T04 — Complex scene	24/24	24/24	All eight bundled criteria judged satisfied
Total	**196/198**	**198/198**	**99.0% vs 100%**

4.1 Detailed interpretation of Test 2

Trial	OpenAI shape	OpenAI color	OpenAI spatial	OpenAI total	Gemini total
S01	3/4	4/4	4/4	11/12	12/12
S02	4/4	4/4	4/4	12/12	12/12
S03	3/4	4/4	4/4	11/12	12/12

The red cube was placed on the pyramid tip as requested, so its **spatial** point was retained. However, OpenAI S01 and S03 appeared to introduce an unnatural pointed feature at the cube–pyramid contact, changing the expected geometric shape. Therefore we deducted **one shape point** for each affected image.

5. Visual Comparison and Discussion

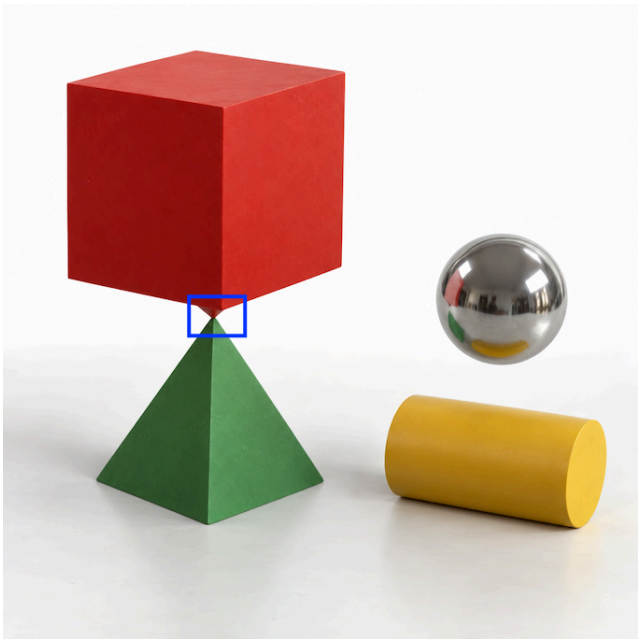
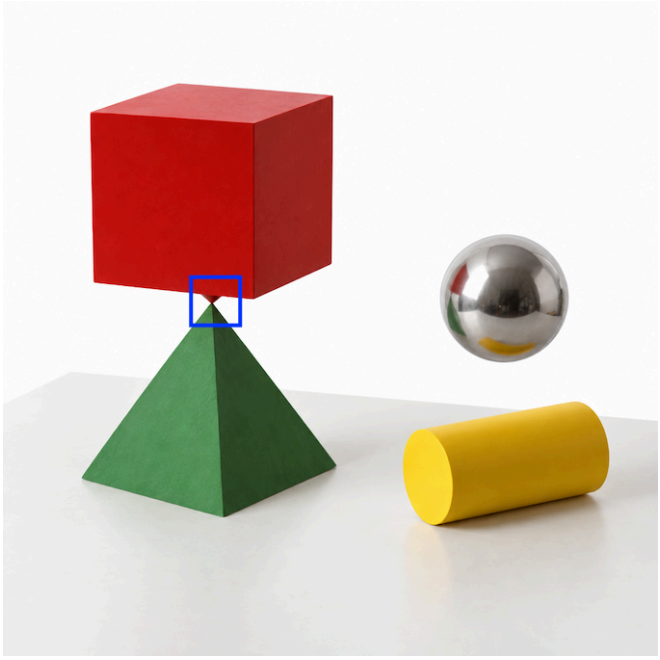
5.1 T01 — Object count, shape, and color

Both tools generated the required three metallic silver spheres, two green pyramids, and four yellow cubes. All six images earned 18/18. The metallic spheres show convincing highlights and reflections, contributing to their apparent realism. On this relatively simple task, the results do not differentiate the tools.

5.2 T02 — Spatial composition versus geometric integrity

Both tools represented an intentionally unusual arrangement: a red cube balancing on a green pyramid, a floating metallic sphere to the right, and a horizontal yellow cylinder below the sphere. Because the prompt explicitly required an implausible arrangement, physical impossibility itself was **not** penalized.

The important distinction was geometric integrity. In OpenAI S01 and S03, an extra pointed contact feature appeared between the cube and pyramid. The object positions were broadly correct, but the cube's shape was less faithful to the requested object. Gemini preserved a more conventional cube geometry at this contact in the three observed trials. This suggests that **approximate spatial compliance and preservation of object shape can diverge** when the prompt asks for an unusual configuration.

	
T02-openai-S01	*T02-openai-S03*

We also noted that the metallic sphere in OpenAI S02 did not visibly reflect the nearby cube and pyramid, while Gemini's sphere reflections were less clearly legible than those in T01. These observations affect perceived realism, but a few rendered images is insufficient to prove a reflection is physically incorrect: viewpoint, roughness, illumination, and the sphere's curvature also affect what can be seen. No reflection score was deducted.

5.3 T03 — Exact text versus camera viewpoint

All six departure-board images reproduced the four headings and 24 flight-data fields correctly, earning 28/28 per image. This is a strong result for the **specific text supplied** in the prompt.

Nevertheless, the models varied in composition. OpenAI S01 and S02 appeared less convincing as views from a standing passenger looking upward at a board mounted high above the airport floor; OpenAI S03 better conveyed the intended angle. Gemini S02 and S03 gave a particularly convincing sense of the board's placement in an airport environment. These differences illustrate why **text accuracy and photographic viewpoint should be assessed separately**. The 28-point rubric captures the former but not the latter.

5.4 T04 — Complex scene, lighting, and aesthetics

Both models produced coherent, broadly photorealistic minimalist living rooms with the specified beige sofa, wooden coffee table, paired blue armchairs, floor lamp, potted plant, and beach view. Natural-looking materials, lighting, and furniture proportions contributed to the visual quality. All six images received 8/8 under the original bundled rubric.

The outputs were not identical in aesthetic treatment. OpenAI added framed wall artwork in some images, and both models included small decorative objects on coffee tables. Gemini S02 included a curtain. These additions were accepted because the prompt prohibited **additional furniture**, not every decorative object. The resulting rooms remained consistent with the broad minimalist aesthetic, although the degree of minimalism differed.

The prompt also requested an **open** floor-to-ceiling sliding door on the left wall and sunlight casting shadows to the right. The outdoor view and illumination are generally convincing, but the precise door-panel position and physical consistency of every shadow are not unambiguously established in all images. The high T04 scores should therefore be read alongside these qualitative caveats.

5.5 Style, aesthetics, realism, and visual quality

Across the four tests, both models created polished images with clear subjects and generally convincing materials. For simple geometric scenes, there was little visible difference in success. For the airport board, Gemini's more contextual compositions in some trials better matched the requested passenger viewpoint, whereas OpenAI often emphasized a large, legible board. For the living-room scenes, OpenAI sometimes introduced more decorative detail, while Gemini's compositions often appeared more restrained. These are **observations about this sample**, not fixed artistic traits of either model.

Claims about which image is “more realistic” should therefore remain qualitative and tied to identifiable visual features rather than expressed as objective superiority.

6. Strengths and Weaknesses of Each Tool

Tool	Strengths observed in this experiment	Weaknesses or qualifications observed
OpenAI	Correct object counts and colors; accurate departure-board text; coherent complex scenes; successful unusual spatial arrangement	Two instances of altered cube contact geometry; some departure-board viewpoints less convincing
Gemini	Correct object counts and colors; accurate departure-board text; consistent cube geometry in T02; convincing airport context in selected T03 images; coherent complex scenes	Some metallic reflections less clearly interpretable than T01; added a curtain in one T04 image

7. Preferred Tool and Intended Future Use

For prompt-sensitive image generation similar to the four tests in this study, Gemini (Nano Banana Pro) is my provisional choice. It matched OpenAI on the three tests where both obtained perfect rubric scores and avoided the two cube-geometry errors observed in T02. Its airport-board compositions also appeared particularly convincing in some trials.

This preference is **conditional**. The numerical difference is only two points out of 198, from a small set of images assessed by one person. For applications prioritizing developer integration, cost, speed, editing, or other styles of artwork, the preferred tool could change. Those factors were not evaluated here.

8. Missing Capabilities and Desired Improvements

The current models are already very useful. The following are my humble opinions on what can be further improved.

1. **Stronger geometric integrity:** Maintain an object’s intended shape even when asked to place it in an unusual spatial relationship, rather than introducing unwanted contact geometry.
2. **Improved physical consistency:** More dependable reflections, contact geometry, lighting direction, and shadows when realistic scenes contain interacting objects.
3. **Better evaluation tools:** Tools that make it easier to compare multiple generations against a checklist of prompt requirements.

These are proposed improvements motivated by the observed examples and workflow; the study does not establish that either provider lacks every capability listed.

9. Limitations of the Evaluation

- **Small sample:** Only three images per model per test, giving 12 images per model. A few errors can materially affect percentages.
- **Comparability of Models:**This exercise does not guarantee that both models are equal in its capabilities.
- **Narrow rubric:** Some camera, physical-realism, and negative constraints were not separately scored. T04 uses bundled criteria.
- **Confounded complexity:** The tests differ in content and task type as well as prompt length and difficulty, preventing a causal conclusion about complexity alone.

10. Conclusion

This controlled comparison found high prompt adherence from both OpenAI and Gemini image-generation models. Each correctly generated the requested object inventory, accurately reproduced the scored airport-board text, and produced coherent photorealistic living-room scenes. The clearest numerical difference occurred in T02, where two OpenAI generations introduced an unnatural pointed feature at the cube–pyramid contact. Scoring this as a **shape-preservation** error.

The broader finding is that **prompt adherence has several dimensions**. A model may satisfy counts, colors, and approximate placement while still producing imperfect geometry, viewpoint, or physical realism. On the selected tests, Gemini is the tentative preferred tool, but the small sample and subjective evaluation limit the strength of that conclusion.

Appendix A

Detailed Test Result

T01 - Counting and Object Generation

Prompt

```
Generate an image containing exactly the following objects:

- 3 silver spheres with metallic surfaces.
- 2 green pyramids.
- 4 yellow cubes.

Arrange all 9 objects on a plain white table.
All objects must be fully visible, clearly separated,
and not overlapping.

Do not include any additional objects or text.
```

Provider: OpenAI

Model: gpt-image-2.5-sunburst

Results:

Image 1:

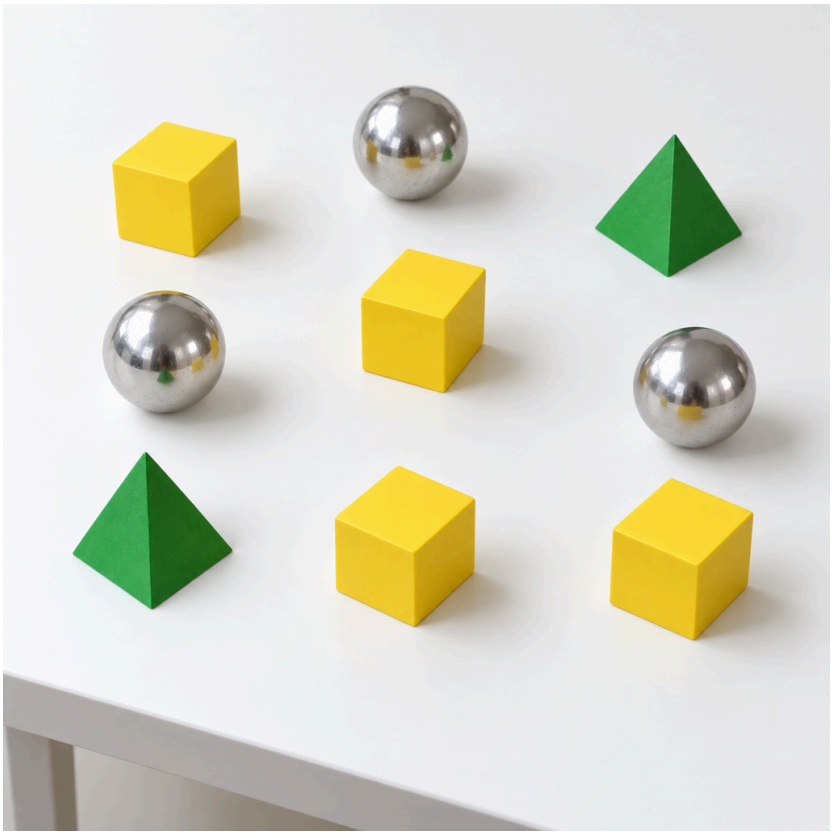


Image 2:

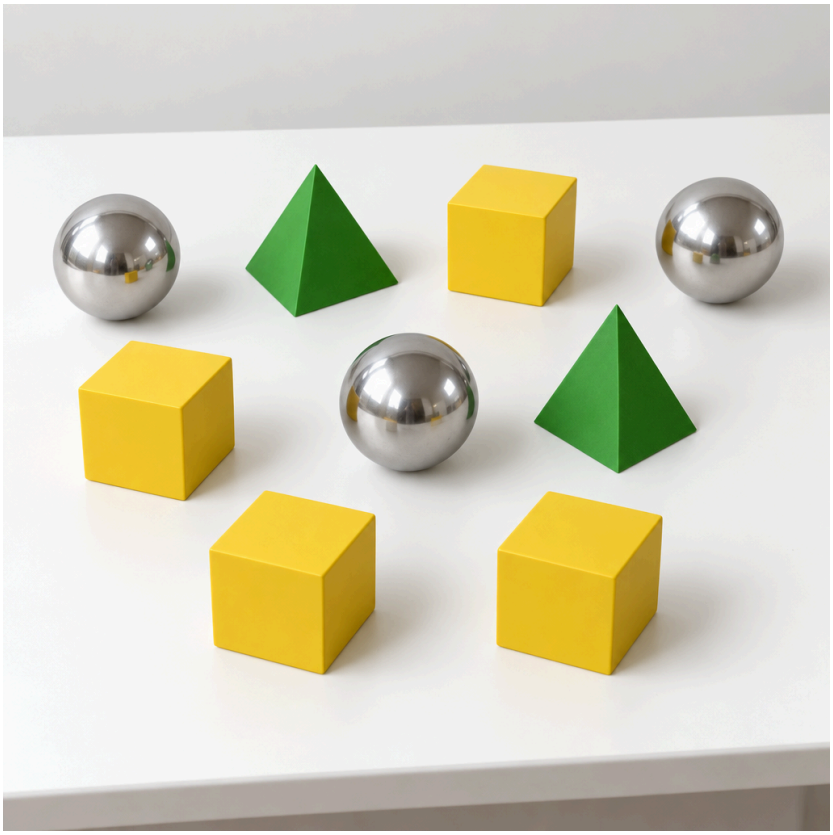
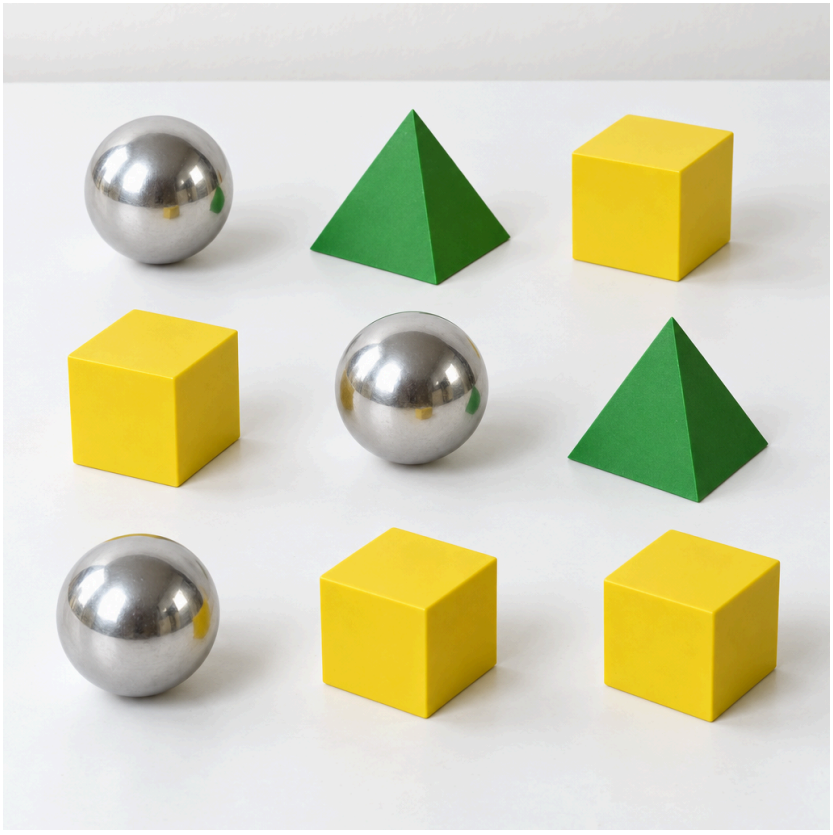


Image 3:



Score Image - T01_openai_S01

- Object Shape: 9/9
- Color: 9/9

Total Score for T01_openai_S01: 18/18

Score Image - T01_openai_S02

- Object Shape: 9/9
- Color: 9/9

Total Score for T01_openai_S02: 18/18

Score Image - T01_openai_S03

- Object Shape: 9/9
- Color: 9/9

Total Score for T01_openai_S03: 18/18

Additional Observation

All images show visible highlights and reflections on the metallic spheres, contributing to perceived realism.

Provider: Gemini

Model: gemini-3-pro-image

Results:

Image 1:

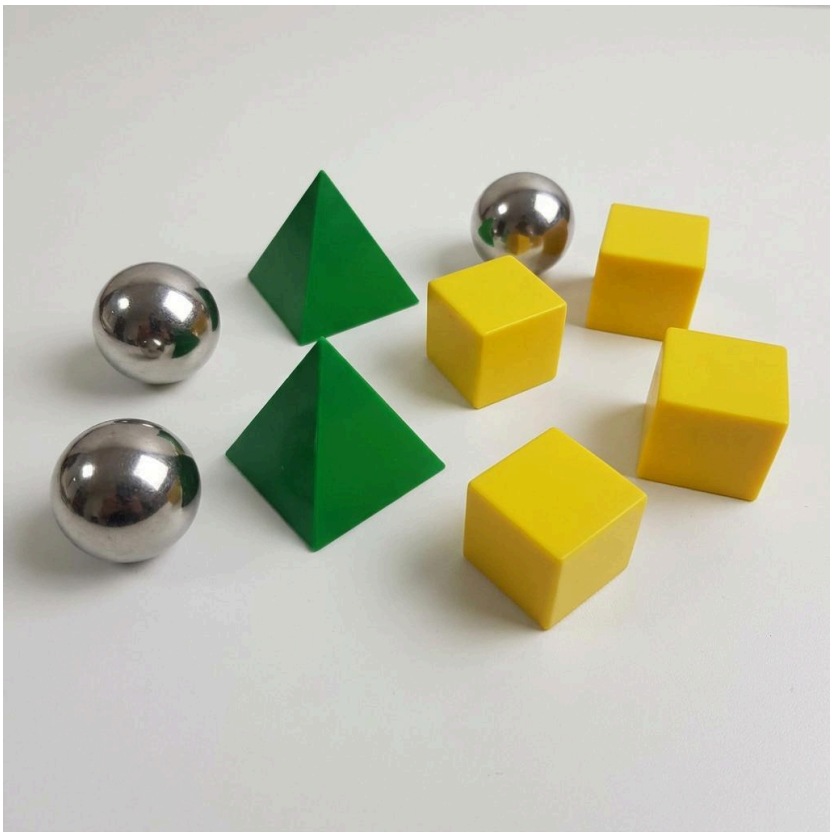
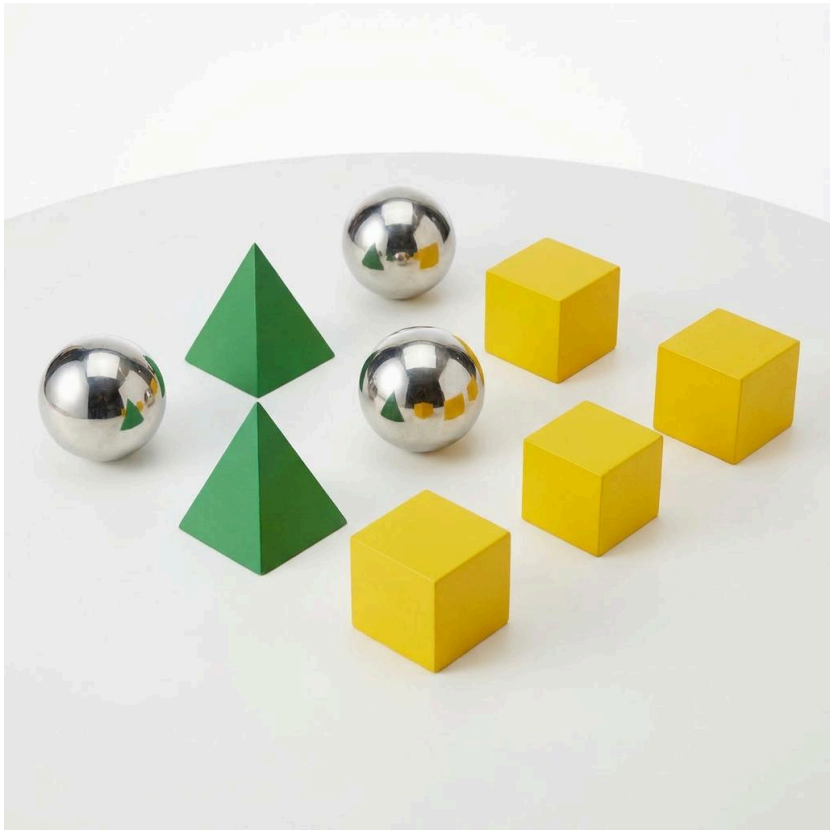


Image 2:

Image 3:



Score Image - T01_gemini_S01

- Object Shape: 9/9
- Color: 9/9

Total Score for T01_gemini_S01: 18/18

Score Image - T01_gemini_S02

- Object Shape: 9/9
- Color: 9/9

Total Score for T01_gemini_S02: 18/18

Score Image - T01_gemini_S03

- Object Shape: 9/9
- Color: 9/9

Total Score for T01_gemini_S03: 18/18

Additional Observation

All images show visible highlights and reflections on the metallic spheres, contributing to perceived realism.

T02 - Spatial Relationship between objects

Prompt

```
Generate a photorealistic image showing the following
arrangement of objects:

- A large red cube balancing directly on the sharp tip
  of a green pyramid.
- The green pyramid is standing upright on a flat white table.
- A metallic sphere is floating in mid-air to the right
  of the pyramid, without any visible support.
- A yellow cylinder is positioned horizontally underneath
  the floating metallic sphere, without touching it.

All objects must be clearly visible.
Do not include any supporting structures, strings,
or additional objects.

Use a slightly elevated camera angle and a plain
white background.
```

Provider: OpenAI

Model: gpt-image-2.5-sunburst

Results:

Image 1:

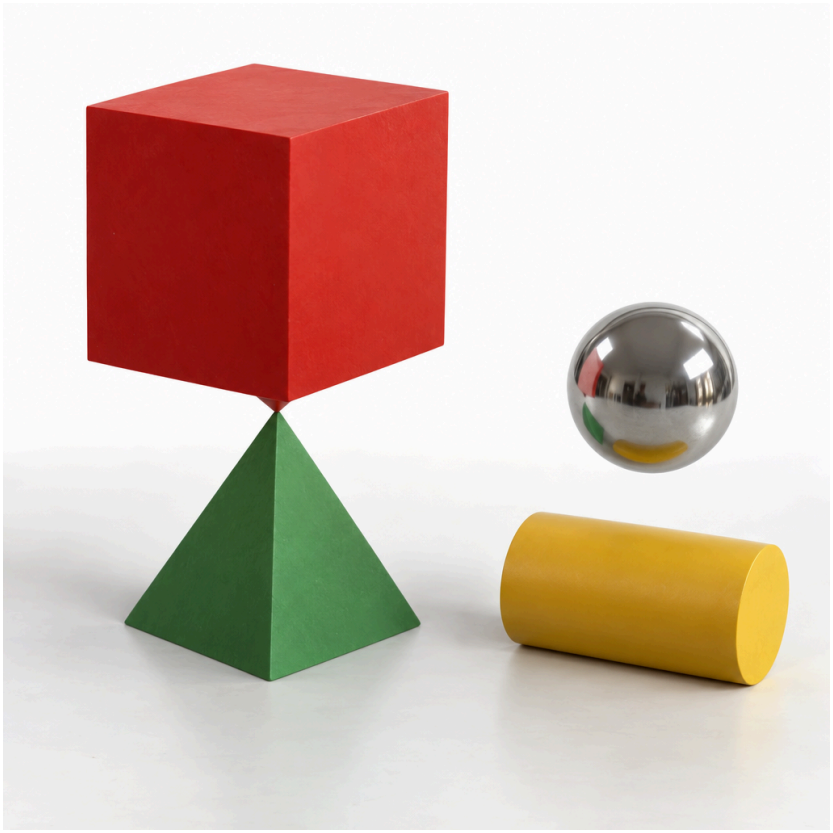


Image 2:

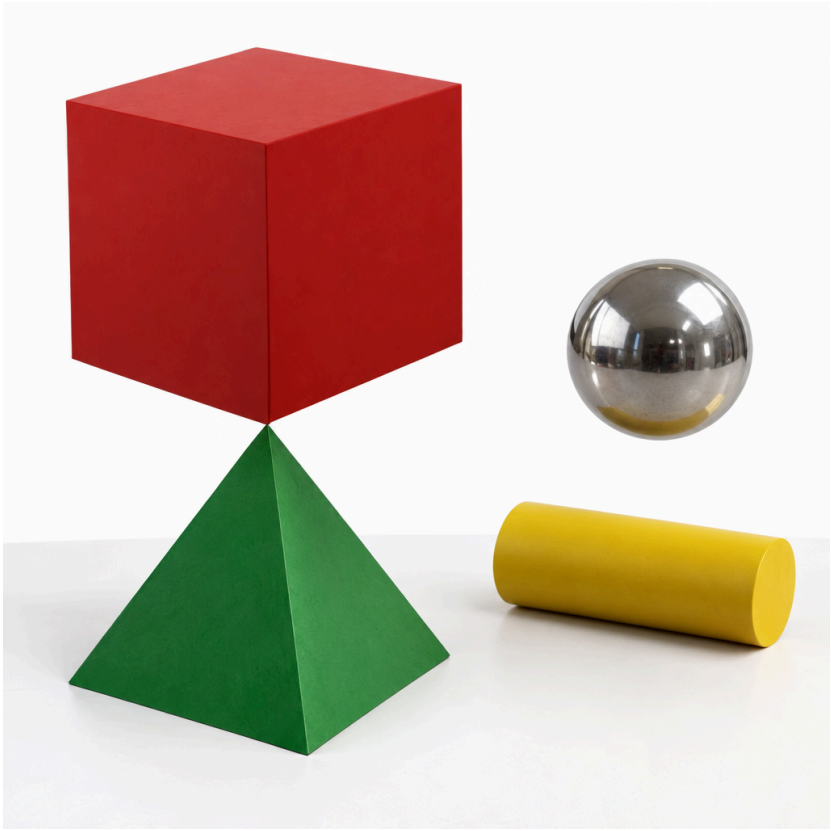
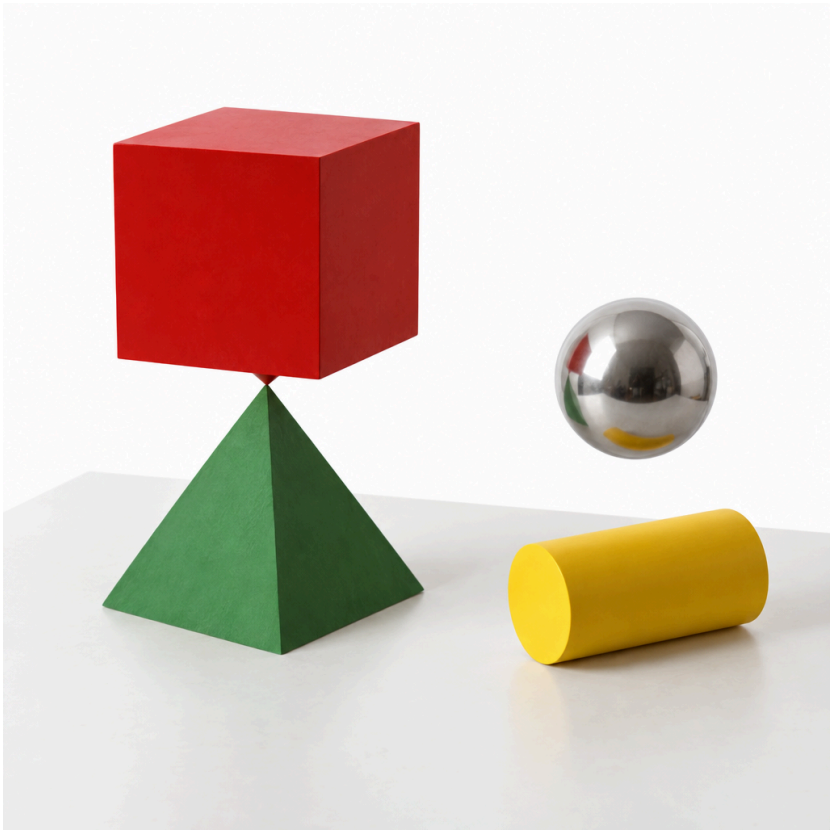


Image 3:



Score Image - T02_openai_S01

- Object Shape: 3/4
- Color: 4/4
- Spatial: 4/4

Total Score for T02_openai_S01: 11/12

Score Image - T02_openai_S02

- Object Shape: 4/4
- Color: 4/4
- Spatial: 4/4

Total Score for T02_openai_S02: 12/12

Score Image - T02_openai_S03

- Object Shape: 3/4
- Color: 4/4
- Spatial: 4/4

Total Score for T02_openai_S03: 11/12

Additional Observation

For the instruction “A large red cube balancing directly on the sharp tip of a green pyramid,” images T02_openai_S01 and T02_openai_S03 generated an unnatural pointed protrusion at the contact point between the cube and pyramid. Rather than preserving the cube’s original geometry and positioning it directly on the pyramid tip, the model appeared to introduce additional geometry to establish the connection. This was considered a failure in geometric accuracy, resulting in a deduction of one point from the shape score for each affected image.

In T02_openai_S02, the sphere does not show readily identifiable reflections of the red cube or green pyramid. This may reduce perceived realism, although the image alone does not establish that the reflections are physically incorrect. Reflections were not part of the numerical rubric.

Provider: Gemini

Model: gemini-3-pro-image

Results:

Image 1:

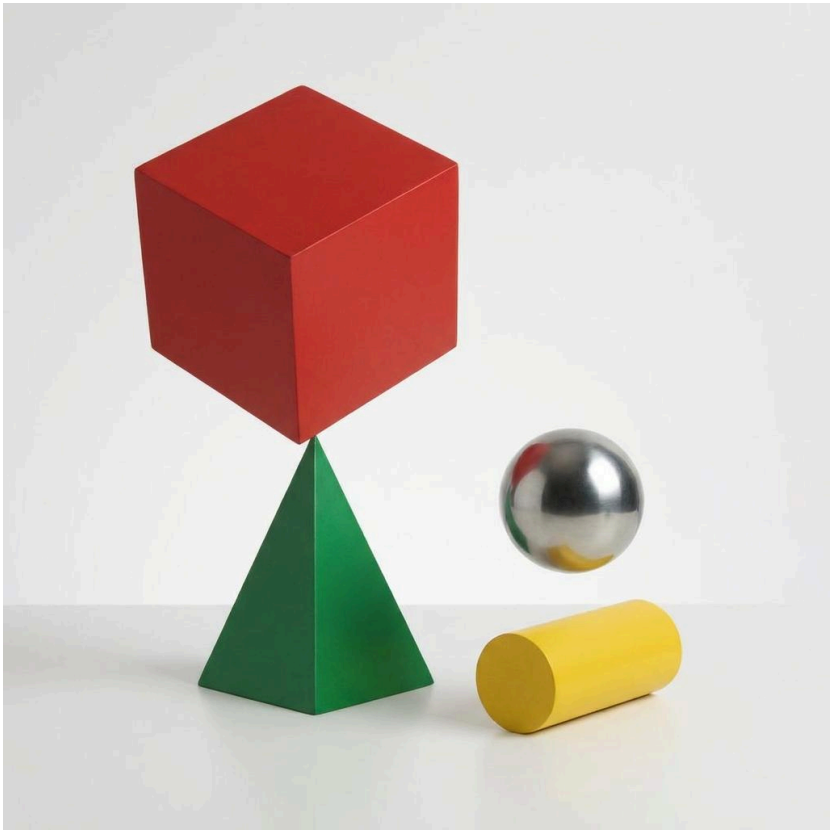
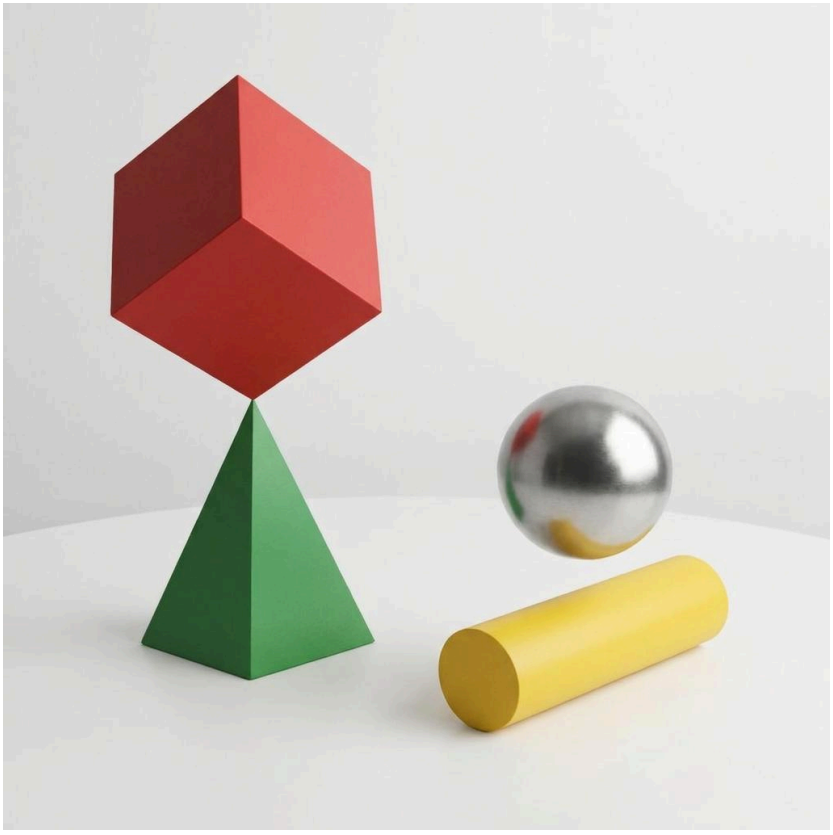


Image 2:

Image 3:



Score Image - T02_gemini_S01

- Object Shape: 4/4
- Color: 4/4
- Spatial: 4/4

Total Score for T02_gemini_S01: 12/12

Score Image - T02_gemini_S02

- Object Shape: 4/4
- Color: 4/4
- Spatial: 4/4

Total Score for T02_gemini_S02: 12/12

Score Image - T02_gemini_S03

- Object Shape: 4/4
- Color: 4/4
- Spatial: 4/4

Total Score for T02_gemini_S03: 12/12

Additional Observation All three images show a reflective metallic surface, although the surrounding scene is less clearly recognizable in the reflections than in Test 1. This is a qualitative observation, not a scored criterion.

T03 - Generate Image with Text Information

Prompt

```
Generate a photorealistic image of a large electronic
flight departure board inside a modern international airport.

The board must display exactly six flights in a table
with four columns labelled:
FLIGHT, DESTINATION, TIME, and STATUS.

Display the following information exactly:

FLIGHT | DESTINATION | TIME | STATUS
SQ318  | LONDON      | 09:45 | BOARDING
JL712  | TOKYO       | 10:20 | ON TIME
EK355  | DUBAI       | 11:05 | DELAYED
QF082  | SYDNEY      | 12:30 | GATE OPEN
CX650  | HONG KONG   | 13:15 | ON TIME
TG402  | BANGKOK     | 14:50 | CANCELLED

All six rows must be present in the specified order.
Every flight number, destination, time, and status
must match exactly.

Mount the departure board high above the airport floor.
Position the camera at a standing adult's eye level,
approximately 1.6 metres above the floor, looking
upward at the board at a 15-degree angle.

Use a natural passenger viewpoint with realistic
perspective distortion.

The entire board must remain visible within the frame.
All text must be sharp, clearly readable, and correctly
aligned in four columns despite the viewing angle.

Use realistic electronic display lighting.

Do not add extra flights, columns, or information
to the board.
```

Provider: OpenAI

Model: gpt-image-2.5-sunburst

Results:

Image 1:

Image 2:

Image 3:

Score Image - T03_openai_S01

- Text generation: 28/28

Total Score for T03_openai_S01: 28/28

Score Image - T03_openai_S02

- Text generation: 28/28

Total Score for T03_openai_S02: 28/28

Score Image - T03_openai_S03

- Text generation: 28/28

Total Score for T03_openai_S03: 28/28

Additional Observation All three images present the requested text clearly. T03_openai_S01 and T03_openai_S02 appear less convincing than T03_openai_S03 in their depiction of the requested upward-looking passenger viewpoint.

Provider: Gemini

Model: gemini-3-pro-image

Results:

Image 1:

Image 2:

Image 3:

Score Image - T03_gemini_S01

- Text generation: 28/28

Total Score for T03_gemini_S01: 28/28

Score Image - T03_gemini_S02

- Text generation: 28/28

Total Score for T03_gemini_S02: 28/28

Score Image - T03_gemini_S03

- Text generation: 28/28

Total Score for T03_gemini_S03: 28/28

Additional Observation All three images present the requested text clearly. T03_gemini_S02 and T03_gemini_S03 provide a particularly convincing airport context and camera perspective.

T04 - Generate Complex Image

Prompt

```
Generate a photorealistic image of a modern minimalist
living room with the following furniture arrangement:

- One beige three-seater sofa positioned against the back wall.
- One rectangular wooden coffee table directly in front of the sofa.
- Two identical blue armchairs, one on each side of the
  coffee table, facing inward.
- One black floor lamp positioned to the right of the sofa.
- One green potted plant positioned to the left of the sofa.
- One large ceiling to floor sliding door on the left wall with the door
open.

Soft morning sunlight enters through the glass from
the left, illuminating the furniture and casting
realistic, consistent shadows toward the right.

The background outside the sliding glass door should show a clear blue sky
with ocean waves and a sandy beach.

Use a wide-angle, eye-level camera perspective.

The room should have natural materials, realistic
fabric textures, accurate furniture proportions,
and a warm, inviting atmosphere.

All furniture must be clearly visible.
Do not include people, additional furniture, or text.
```

Provider: OpenAI

Model: gpt-image-2.5-sunburst

Results:

Image 1:

Image 2:

Image 3:

Score Image - T04_openai_S01

- Satisfied Prompt Demand: 8/8

Total Score for T04_openai_S01: 8/8

Score Image - T04_openai_S02

- Satisfied Prompt Demand: 8/8

Total Score for T04_openai_S02: 8/8

Score Image - T04_openai_S03

- Satisfied Prompt Demand: 8/8

Total Score for T04_openai_S03: 8/8

Additional Observation Additional object was placed on the coffee table and 2 of the images added a painted picture. Since we do not have strict constraint on wall painting and coffee table objects. We consider the addition acceptable.

All three images appear broadly photorealistic; the precise open state of the sliding door is not always unambiguous.

Provider: Gemini

Model: gemini-3-pro-image

Results:

Image 1:

Image 2:

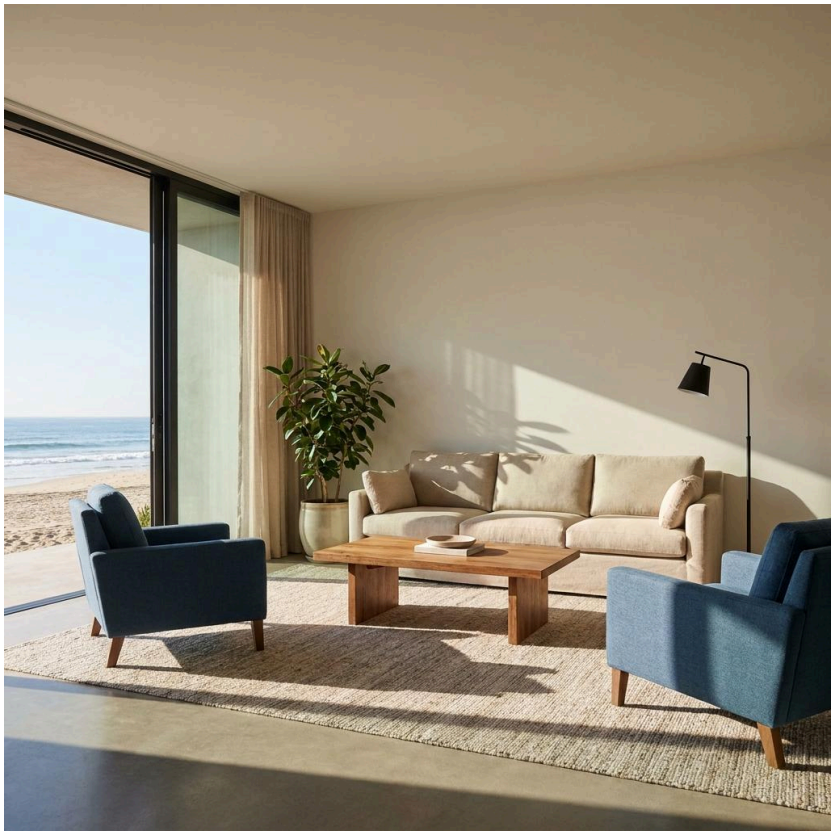


Image 3:

Score Image - T04_gemini_S01

- Satisfied Prompt Demand: 8/8

Total Score for T04_gemini_S01: 8/8

Score Image - T04_gemini_S02

- Satisfied Prompt Demand: 8/8

Total Score for T04_gemini_S02: 8/8

Score Image - T04_gemini_S03

- Satisfied Prompt Demand: 8/8

Total Score for T04_gemini_S03: 8/8

Additional Observation Additional small decorative objects appear on the coffee table; these were accepted because the prompt prohibited additional furniture, not all decorative items. T04_gemini_S02 also includes a curtain, which was not treated as additional furniture.

All three images appear broadly photorealistic; the precise open state of the sliding door is not always unambiguous.

END